# Arithmetic Sequences

Exam code: 9MA0

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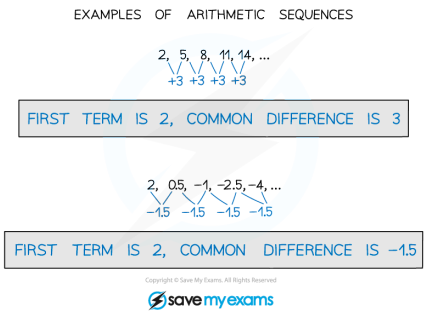
## Arithmetic sequences

### What do I need to know about arithmetic sequences?

- In an **arithmetic sequence**, the difference between consecutive terms in the sequence is constant




- That constant difference is known as the **common difference** of the sequence
- You need to know the ***n*th term formula** for an arithmetic sequence

$$u_n = a + (n-1)d$$

  - $a$ is the first term
  - $d$ is the common difference

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

## Examiner Tips and Tricks

If you know two terms in an arithmetic sequence you can find ***a*** and ***d*** using simultaneous equations.

e.g. THE $15^{th}$ TERM OF AN ARITHMETIC SEQUENCE IS $-19$.
THE $21^{st}$ TERM IS $-28$.
FIND THE FIRST TERM AND THE COMMON DIFFERENCE.

$u_n = a + (n-1)d$

$a + 14d = -19$ (1)
$a + 20d = -28$ (2)

SOLVE THESE SIMULTANEOUS EQUATIONS

$6d = -9$
$d = -1.5$

(2) MINUS (1)

$a + 14(-1.5) = -19$
$a = 2$

SUBSTITUTE INTO (1)

THE FIRST TERM IS 2 AND
THE COMMON DIFFERENCE IS $-1.5$

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## Worked Example

The first three terms in an arithmetic sequence are $12, k^2, 5k, \ldots$

Find two possible values of $k$.

THIS IS AN ARITHMETIC SEQUENCE SO THERE'S A SINGLE COMMON DIFFERENCE

$d = k^2 - 12$ — $2^{nd}$ TERM MINUS $1^{st}$ TERM

AND

$d = 5k - k^2$ — $3^{rd}$ TERM MINUS $2^{nd}$ TERM

SO $k^2 - 12 = 5k - k^2$

$2k^2 - 5k - 12 = 0$

$(2k+3)(k-4) = 0$ — SOLVE THE QUADRATIC

$k = -\frac{3}{2}$ OR $k = 4$

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